

Evaluation Report: The Tusheti Community Broadband Rollout—Technical Resilience and Stakeholder Synergy

1. Strategic Project Context and Multi-Stakeholder Governance

The Tusheti region, situated on the northern slopes of the Greater Caucasus Mountains, represents one of Georgia's most geographically isolated territories. Bordering the autonomous republics of Chechnya and Dagestan, the region's extreme terrain and sparse population historically excluded it from national fiber-optic expansion plans. Recognizing this digital divide, the Tusheti Community Broadband project was launched in October 2016 to establish a fixed wireless network using high-capacity backhaul. This infrastructure was a strategic necessity to bridge the gap in economic sustainability, education, and healthcare for a region where traditional telecommunications were non-existent.

The project's viability was anchored in a 2016 Memorandum of Understanding (MoU) that integrated global expertise with local operational capacity through six primary stakeholders:

- **Internet Society (ISOC):** Provided the primary grant funding and global strategic oversight.
- **ISOC Georgia Chapter:** Managed project monitoring, technical training, and advisory services for tariffication.
- **Small and Medium Telecom Operators Association of Georgia (SMOAG):** Spearheaded hardware evaluation, procurement coordination, and overall project management.
- **LTD Freenet:** Served as the primary technical partner, providing transit services and essential infrastructure access.
- **Internet Development Initiative (IDI):** Tasked with content creation and service development to ensure the network translated into tangible economic gain.
- **Tusheti Development Fund (TDF):** The local beneficiary and community-based operator responsible for long-term ownership and maintenance.

A critical success factor was the "Contribution" framework described in the MoU. **LTD Freenet** provided a vital operational subsidy by offering "free of charge" colocation on its Ruispiri mast—including space, security, and uninterruptible power. Additionally, Freenet provided 100 Mbps of global/local transit without fees, a contribution valued at approximately 3,200 GEL per month. To further ensure viability, the technical team leveraged a specific **5.4%–5.6% discount** from the supplier **Intellcom Group** for all hardware procurement.

2. Network Topology Evolution: Theoretical Design vs. Field Reality

In high-altitude deployments, the disparity between theoretical radio propagation models and environmental reality can lead to catastrophic system failure. The Tusheti project required a radical

departure from the initial designs provided by the Georgian National Communications Commission (GNCC) and the National Regulatory Authority (NRA) to account for the unique alpine microclimate and Non-Line-of-Sight (NLOS) challenges.

Feature	Initial Theoretical Approach (NRA/GNCC)	Final Implementation Reality (Field-Corrected)
Mast Placement	Lower altitudes near village centers.	High-altitude ridges (Diklo Mountain, Abano Pass).
Backhaul Tech	Standard P2P bridges.	AirFiber high-capacity backhaul technology.
Coverage Strategy	Standard 90° sectoral coverage.	Wide-angle 240° coverage for convoluted topography.
Sector Hardware	7 units of AM-5G19-120.	14 units of Ubiquiti LBE-5AC-16-120.
Signal Delivery	Short-range village-to-village hops.	High-altitude "bridge points" for wide-area LOS.

Critical to this evolution were the local community meetings in **Alvani**. These consultations provided "field-truth" data that corrected theoretical mast placements, specifically identifying "**snow slips**" (**avalanches**) that would have destroyed masts placed according to the original GNCC scheme. The technical team identified three primary environmental barriers:

- 1. Extreme Snow Accumulation:** Surveys revealed snow depths reaching 1.5 meters, threatening mast stability and physically obstructing Fresnel zones at lower altitudes.
- 2. Vegetation/NLOS Obstruction:** Unforeseen evergreen growth at lower altitudes created signal barriers that theoretical models failed to capture.
- 3. Ultra-Standard Lightning Risks:** Key transmission points possessed high sulfide and ferrite metal content, attracting catastrophic strikes. The case study of the mobile operator **Mobitel** served as a warning; despite grounding masts to **ISO 9001:2008** standards, Mobitel lost thousands of dollars in equipment and eventually suspended operations in the region.

3. Hardware Adaptation and Infrastructure Resilience

To ensure 12-month stability, standardized ICT hardware required bespoke modification for the Caucasus environment. Technical leads Konstantin Stalinsky and Murad Charelidze shifted from the original radio specifications to a more robust, wide-angle configuration.

- **Radio Equipment Shift:** The team replaced the original 90° sectoral antennas with **Ubiquiti LBE-5AC-16-120** models. This move doubled the unit count to 14, increasing coverage angles to 240°. This strategic pivot allowed a single mast to service multiple settlements (such as Kumelaurta and Indurta) that were previously "invisible" to the network.
- **Solar Power Reconfiguration:** Initial 24V setups with two 100W panels proved insufficient due to winter azimuth efficiency losses. Calculations showed that changing the panel angle to the required **53°** on a mountaintop was physically impossible. The only solution was to increase the quantity of panels to **four (200W total)** per mast.
- **Power Management:** The system was reconfigured using the **POE-BOX8-G**, ensuring the power controller could simultaneously feed the network and charge the accumulators during the short daylight hours of the alpine winter.
- **Environmental Hardening:** To survive the geological lightning risks, the team integrated **Andeli thunder protection systems** and custom-manufactured mast mounts designed in Tbilisi and Telavi to withstand the mechanical stress of high-altitude "snow slips."

4. The Community-Based ISP Model: A Sustainability Analysis

The Tusheti project utilizes a non-profit, community-managed business model that departs from traditional profit-driven telecommunications. The **Tusheti Development Fund (TDF)** operates the network with a strict mandate: profits cannot exceed **10%** of operational and capital expenditures. All surplus revenue is legally required to be reinvested into network maintenance, upgrades, and expansion.

Social Equity and Service Mandate: Section 2.3 of the MoU establishes a "No Refusal" mandate. The TDF is obligated to provide service to any point in Tusheti where technical feasibility exists, ensuring that isolated private households have the same digital rights as large commercial guest houses.

Subscriber Impact and Business Model: The network currently delivers packet speeds of **3-5 Mbps** to a diverse subscriber base. The financial entry point was designed to be accessible for the local community:

- **Installation Cost:** 100–120 GEL (including CPE).
- **Monthly Subscription:** 25–30 GEL.

Current "wifitusheti metushael" data shows the network's reach:

- **SMEs & Tourism:** Guest houses including "Tusheti Travel" (Soso Babulaidze), "Samtsikhe" (Dartlo), and "Shina" (Zemo Omalo).
- **State Agencies:** Critical support for the **Border Guard Authority** and the **Protected Areas Agency** in Omalo.
- **Private Residents:** Individuals like **Soso Bakuriadze** (Chigho) and **Irakli Khvedaguridze** (Bochora) are now integrated into the digital economy.

5. Strategic Template for Remote Connectivity: Conclusions and Scalability

The Tusheti project serves as a scalable blueprint for global "Internet Society" initiatives. The project demonstrates that technical resilience is as much about community synergy as it is about hardware, as evidenced by the Alvani meetings which prevented the loss of infrastructure to seasonal avalanches.

Backhaul and the Khevsureti Extension: The network topology was pre-designed to serve as a backhaul for neighboring regions. The high-altitude hubs at **Abano Pass and Diklo Mountain** were specifically chosen to provide the line-of-sight required for the "Khevsureti Extension." Utilizing **AirFiber** technology, the current backbone is ready to transmit signals to the neighboring Khevsureti region, which faces identical geographic barriers.

Core Principles for Future Remote Initiatives:

1. **Technical Flexibility:** Infrastructure must be "field-corrected" to account for micro-environmental factors (e.g., 53° solar constraints and soil conductivity).
2. **Community Ownership:** Locating ownership within a local non-profit fund (TDF) ensures reinvestment and long-term maintenance.
3. **Multi-Sectoral Subsidy:** Leveraging existing infrastructure from partners like LTD Freenet is essential to overcoming the high CAPEX of mountain deployments.

The Tusheti Community Broadband project has successfully transformed high-altitude connectivity from a theoretical goal into a community-managed utility, proving that even the most convoluted topography can be bridged through technical adaptation and stakeholder synergy.